

Titer Determination of NaOH 0.1 mol/L

Method for the standardization of 0.1mol/L sodium hydroxide using potassium hydrogen phthalate as the primary standard.

Sample	Primary standard:- Potassium hydrogen phthalate (80 – 120 mg)
Compound	Potassium hydrogen phthalate, KHP, M =204.23 g/mol, z = 1
Chemicals	50mL deionized water
Titrant	Sodium hydroxide, NaOH c(NaOH)=0.1 mol/L
Standard	--
Indication	DGi111-SC combined pH glass electrode (MT-51109500)
Chemistry	$\text{NaOH} + \text{HOOC-C}_6\text{H}_4\text{-COOH} \rightarrow \text{Na}^+ + \text{K}^+ + ^-\text{OOC-C}_6\text{H}_4\text{-COO}^-$
Calculation	$R = m/(\text{VEQ} \cdot c \cdot C)$ $C = M/(10 \cdot p \cdot z)$ VEQ=Titrant consumption in mL c = Concentration of the titrant. m = mass of the sample in g. M = Molar mass of the sample p = Purity of the standard. z = Equivalent number
Waste disposal	Neutralization before final disposal as aqueous solution
Author, Version	Vineesh Pallath, MSG AnaChem, Version 2.0 Revised: Cosimo De Caro, MSG AnaChem

Preparation and Procedures

CAUTION

- Use safety goggles, a lab coat and wear gloves.
- Ensure accurate cleaning of sensor is sufficient after each titration.

Preparation of 0.1 mol/L NaOH:

- Accurately weigh 4.0 g of NaOH to a 1000mL volumetric flask and dissolved in 500mL of deionized water.
- Heat will generate, cool the solution to room temperature and fill up to 1000mL with deionized water.

Preparation of sample solution:

- Accurately weigh 0.08 – 0.12g of potassium hydrogen phthalate in to a glass titration beaker and add 50mL deionized water and keep the beaker in to Rondolino sample changer.
- Run the method. After completion of titration, the electrode, stirrer and titration tubes will be rinsed by deionized water by means of membrane pump and preceded by condition of electrode with deionized water for 30s in condition beaker placed on Rondolino by selecting 7 on the knob on the top of Rondolino.

Remarks

- The method parameters have been optimized for the sample of this application. It may be necessary to adapt the method to your specific sample.
- Glass beakers are recommended in order to avoid any interference during weighing due to electrostatic force.
- To keep the titer of the of the sodium hydroxide constant, it is important to avoid intake CO₂ from the air which leads to the formation of carbonate. This is achieved by using a suitable adsorbing medium i.e sodium hydroxide on support. It is filled in a drying tube (ME-23961), holder ME-23915 on top of the burette holder.
- This method allows a fully automated analysis procedure. The method can be easily modified for manual operation. Select "Manual stand" in the method function "Titration stand".

Instruments

- Titration Excellence T50/T70/T90
(Other Titrators: With modifications or manual operations are necessary)
- Rondolino - (ME-51108500)
- 1 x 10mL DV1010 glass burette (ME-51107501),
- XP205 Balance (ME-11106024)

Accessories

- LabX® pro titration software
- Glass titration beaker (ME- 00101446).

Results

All results

Method-ID	VPTiterNaOH
Sample	0.08360 (1/6)
R1 (Titer)	1.00288
Sample	0.08818 (2/6)
R1 (Titer)	1.00148
Sample	0.08465 (3/6)
R1 (Titer)	1.00328
Sample	0.08477 (4/6)
R1 (Titer)	1.00264
Sample	0.08436 (5/6)
R1 (Titer)	1.00110
Sample	0.08426 (6/6)
R1 (Titer)	1.00157

Statistics

Method-ID	VPTiterNaOH
R1	Titer
Samples	6
Mean	1.00216
s	0.00089
srel	0.089%

Titration curve

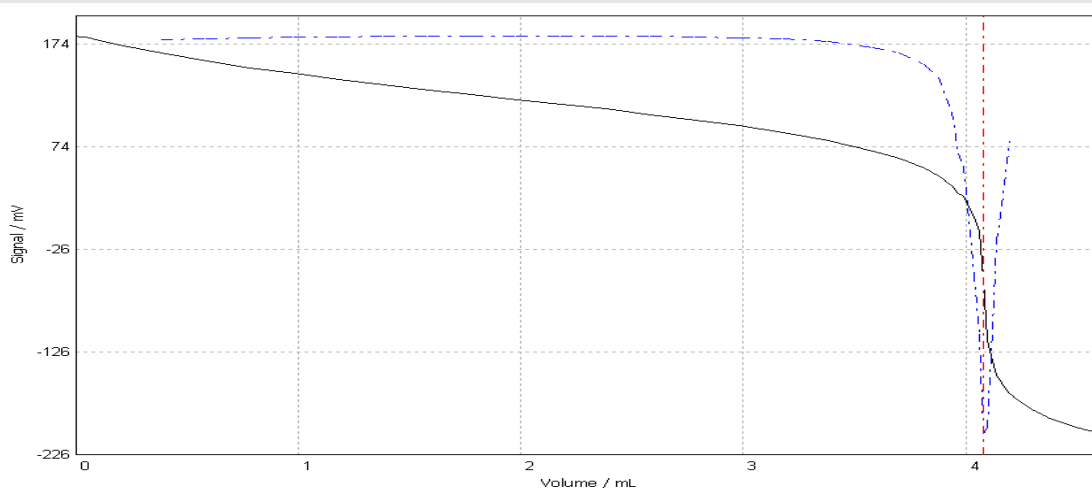


Table of measured values

	Volume mL	Increment mL	Signal mV	Change mV	1st deriv. mV/mL	Time s	Temperature °C
EQP1	0.0000	0.0200	182.1	NaN	NaN	0	25.0
	0.0200	0.0200	181.6	-0.5	NaN	3	25.0
	0.0400	0.0500	180.9	-0.7	NaN	6	25.0
	0.0900	0.1250	178.7	-2.2	NaN	9	25.0
	0.2150	0.1675	172.9	-5.8	NaN	12	25.0
	0.3825	0.2000	166.0	-6.9	-39.53	15	25.0
	0.5825	0.2000	158.2	-7.8	-35.52	18	25.0
	0.7825	0.2000	151.4	-6.8	-31.86	21	25.0
	0.9825	0.2000	145.6	-5.8	-28.96	24	25.0
	1.1825	0.2000	140.1	-5.5	-26.92	27	25.0
	1.3825	0.2000	134.7	-5.4	-25.28	30	25.0
	1.5825	0.2000	129.8	-4.9	-24.18	34	25.0
	1.7825	0.2000	124.9	-4.9	-23.50	36	25.0
	1.9825	0.2000	120.2	-4.7	-23.38	40	25.0
	2.1825	0.2000	115.8	-4.4	-23.53	43	25.0
	2.3825	0.2000	111.0	-4.8	-24.37	46	25.0
	3.9370	0.0580	36.6	-9.6	-308.91	77	25.0
	3.9690	0.0320	28.4	-8.2	-462.14	80	25.0
	3.9890	0.0200	25.6	-2.8	-507.23	83	25.0
	4.0390	0.0500	5.4	-20.2	-910.03	89	25.0
	4.0590	0.0200	-7.9	-13.3	-1134.37	94	25.0
	4.0790	0.0200	-55.1	-47.2	-1506.88	108	25.0
	4.081660	NaN	-62.9	NaN	-1506.89	NaN	25.0
	4.0990	0.0200	-114.1	-59.0	-1478.56	119	25.0
	4.1190	0.0200	-134.6	-20.5	-1112.46	124	25.0
	4.1415	0.0225	-148.0	-13.4	-785.64	127	25.0
	4.3710	0.0745	-189.7	-7.6	NaN	143	25.0
	4.4815	0.1105	-198.3	-8.6	NaN	146	25.0
	4.6175	0.1360	-206.4	-8.1	NaN	149	25.0

Comments

- The mean value of the titer is automatically stored as part of the setup by the function TITER.
- Purity of the primary standard, potassium hydrogen phthalate (p) = 100.00%

Literature:

- Mettler-Toledo Application note M002
- Mettler-Toledo Application note M435
- Mettler-Toledo Application note M521

001 Title

Type	Titer determination
Compatible with	T50 / T70 / T90
ID	VPtiterNaOH
Title	Titer 0.1 mol/L NaOH VP
Author	admin
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002 Sample (Titer)

Titrant	NaOH
Concentration	0.1 mol/L
Standard	Potassium hydrogen phthalate
Type of standard	Solid
Entry type	Weight
Lower limit	0.08 g
Upper limit	0.12 g
Density	1.0 g/mL
Correction factor	1.0
Temperature	25.0°C
Entry	Arbitrary

003 Titration stand (Rondo/Tower A)

Type	Rondolino TTL
Titration stand	Rondolino TTL1
Lid Handling	No

004 Stir

Speed	30 %
Duration	60 s
Condition	No

005 Titration (EQP) [1]**Titrant**

Titrant	NaOH
Concentration	0.1 mol/L

Sensor

Type	pH
Sensor	DG111-SC
Unit	mV

Temperature acquisition

Temperature acquisition	No
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Stir

Speed	30%
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Predispense

Mode	None
Wait time	0 s

Control

Control	User
Titrant addition	Dynamic
dE (set value)	8.0 mV
dV (min)	0.02 mL
dV (max)	0.2 mL
Mode	Equilibrium controlled
dE	1.0 mV
dt	1 s
t (min)	3 s
t (max)	30 s

Evaluation and recognition

Procedure	Standard
Threshold	200.0 mV/mL
Tendency	Negative
Ranges	0
Add. EQP criteria	No

Termination

At Vmax	10.0 mL
At potential	No
At slope	No
After number of recognized EQPs	Yes
Number of EQPs	1
Combined termination criteria	No
Accompanying stating	No
Condition	No
Condition	No

006 Calculation R1

Result	Titer
Result unit	
Formula	$R1 = m / (VEQ \cdot c \cdot C)$
Constant	$C = M / (10 \cdot p \cdot z)$
M	M[KHP]
z	z[KHP]
Decimal places	5
Result limits	No
Record statistics	Yes
Extra statistical func.	No
Send to buffer	No
Condition	No

007 Rinse

Auxiliary reagent	Water
Rinse cycles	1
Vol.per cycle	10mL
Position	Current Position
Drain	No
Condition	No

008 Conditioning

Type	Fix
Interval	1
Position	Conditioning beaker
Time	30s
Speed	50%
Condition	No

009 End of sample**010 Titer**

Titrant	NaOH
Concentration	0.1 mol/L
TITER	Mean[R1]
Limits	No
Conditions	No

011 Calculation R2

Result	Mean Titer
Result unit	--
Formula	$R2 = \text{Mean}[R1]$
Constant	C=1
M	M[KHP]
z	z[KHP]
Decimal places	5
Result limits	No
Record statistics	Yes
Send to buffer	No
Condition	No